LibreNMS Network Monitoring System

Homelab Deployment — Portfolio Documentation

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1. Project Overview

This document covers the complete deployment of LibreNMS on a Rocky Linux 9.7 homelab node. LibreNMS is a free, open-source network management system (NMS) that uses SNMP to monitor network devices, servers, and infrastructure. It provides real-time availability maps, interface traffic graphs, CPU and memory health monitoring, event logging, and alerting.

This deployment demonstrates practical skills in Linux server administration, network monitoring, SNMP configuration across Linux and Windows platforms, database administration, web server configuration, and SELinux policy management — all relevant to SOC analyst and security administrator roles.

2. Lab Environment

2.1 Network Topology

The Topton N150 mini PC acts as a transparent network TAP — all internet traffic flows through it via a Linux bridge (br0), allowing Suricata IDS to inspect traffic in promiscuous mode. LibreNMS is hosted on this same device, making it the central monitoring point for the homelab.

```
Internet → enp1s0 → br0 (bridge) → enp2s0 → WiFi Router WAN
                          |
                    Suricata IDS
                    enp3s0 ← WiFi Router LAN → TAROX PC
```

2.2 Device Inventory

Device	Role	IP Address	OS
Topton N150 Mini PC	Network TAP / IDS / LibreNMS host	192.168.1.217	Rocky Linux 9.7
TAROX PC (NUC7i5BNH)	Windows Server DC / Splunk SIEM	192.168.1.10	Windows Server 2025
EE Smart Hub 7HC25H	WiFi Router / Gateway / DHCP	192.168.1.1	Proprietary

2.3 Topton N150 Specification

Specification	Detail
CPU	Intel N150 (quad-core, 3.6 GHz boost)
RAM	32 GB DDR5 4800 MHz (31 GiB usable)
Storage	1 TB NVMe SSD
OS	Rocky Linux 9.7 (Blue Onyx)
NICs	4x Ethernet (enp1s0, enp2s0, enp3s0, enp4s0)
Bridge	br0 — transparent bridge (WAN-side and LAN-side)
Management	Cockpit — https://192.168.1.217:9090

3. Software Stack

Component	Version	Purpose
LibreNMS	Latest (daily channel)	NMS platform
MariaDB	10.5.29	Database backend
Nginx	1.20.1	Web server
PHP	8.2.30 (Remi repo)	Application runtime
net-snmp	5.9.1	SNMP daemon on Topton
python3-PyMySQL	0.10.1	Python DB connector for poller
python3-dotenv	0.19.2	Environment file loader for poller
fping / mtr / rrdtool	Current	Ping, traceroute, RRD graphing

4. Installation Steps

4.1 Enable Required Repositories

LibreNMS requires PHP 8.2 which is available via the Remi repository. EPEL is needed for additional packages such as fping and ImageMagick.

```
sudo dnf install -y epel-release
sudo dnf install -y https://rpms.remirepo.net/enterprise/remi-release-9.rpm
sudo dnf module reset php -y
sudo dnf module enable php:remi-8.2 -y
```

4.2 Install Dependencies

Install all required packages in separate commands to avoid multi-line continuation issues:

```
sudo dnf install -y git curl unzip
sudo dnf install -y nginx mariadb-server
sudo dnf install -y php php-fpm php-cli php-mysqlnd php-gd php-mbstring php-xml php-zip php-bcmath php-snmp php-curl php-opcache
sudo dnf install -y net-snmp net-snmp-utils fping mtr rrdtool ImageMagick python3
sudo dnf install -y python3-PyMySQL python3-dotenv
```

4.3 Configure MariaDB

```
sudo systemctl start mariadb
sudo systemctl enable mariadb
sudo mysql_secure_installation
```

When prompted during mysql_secure_installation:

- Switch to unix_socket authentication: Y
- Change root password: n
- Remove anonymous users: Y
- Disallow root login remotely: Y
- Remove test database: Y
- Reload privilege tables: Y

Create the LibreNMS database (unix_socket enabled, so use sudo mysql):

```
sudo mysql
CREATE DATABASE librenms CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'librenms'@'localhost' IDENTIFIED BY 'YourStrongPassword!';
GRANT ALL PRIVILEGES ON librenms.* TO 'librenms'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

4.4 Create System User and Clone Repository

```
sudo useradd librenms -d /opt/librenms -M -r -s /usr/bin/bash
sudo git clone https://github.com/librenms/librenms.git /opt/librenms
sudo chown -R librenms:librenms /opt/librenms
sudo chmod 771 /opt/librenms
cd /opt/librenms
sudo -u librenms ./scripts/composer_wrapper.php install --no-dev
```

4.5 Fix Directory Permissions and SELinux

The composer install may run before some directories exist. Create and set permissions explicitly:

```
sudo mkdir -p /opt/librenms/rrd /opt/librenms/logs /opt/librenms/bootstrap/cache
/opt/librenms/storage
sudo chown -R librenms:librenms /opt/librenms
sudo setfacl -d -m g::rwx /opt/librenms/rrd /opt/librenms/logs
/opt/librenms/bootstrap/cache /opt/librenms/storage
sudo setfacl -R -m g::rwx /opt/librenms/rrd /opt/librenms/logs
/opt/librenms/bootstrap/cache /opt/librenms/storage
```

Apply SELinux contexts:

```
sudo semanage fcontext -a -t httpd_sys_rw_content_t '/opt/librenms/logs(/.*)?'
sudo semanage fcontext -a -t httpd_sys_rw_content_t
'/opt/librenms/bootstrap/cache(/.*)?'
sudo semanage fcontext -a -t httpd_sys_rw_content_t '/opt/librenms/storage(/.*)?'
sudo semanage fcontext -a -t httpd_sys_rw_content_t '/opt/librenms/rrd(/.*)?'
sudo restorecon -RFvv /opt/librenms
```

4.6 Configure PHP-FPM

```
sudo cp /etc/php-fpm.d/www.conf /etc/php-fpm.d/www.conf.bak
sudo sed -i 's/^user = apache/user = librenms/' /etc/php-fpm.d/www.conf
sudo sed -i 's/^group = apache/group = librenms/' /etc/php-fpm.d/www.conf
sudo sed -i 's/^listen = .*/listen = \/run\/php-fpm\/librenms.sock/' /etc/php-
fpm.d/www.conf
sudo sed -i 's/^;listen.owner/listen.owner/' /etc/php-fpm.d/www.conf
sudo sed -i 's/^;listen.group/listen.group/' /etc/php-fpm.d/www.conf
sudo sed -i 's/^;listen.mode/listen.mode/' /etc/php-fpm.d/www.conf
```

```
sudo systemctl enable php-fpm && sudo systemctl start php-fpm
```

4.7 Configure Nginx

Create the LibreNMS site configuration at /etc/nginx/conf.d/librenms.conf:

```
server {
    listen 80;
    server_name 192.168.1.217;
    root /opt/librenms/html;
    index index.php;
    charset utf-8;
    gzip on;
    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }
    location ~ [^/]\.php(/|$) {
        fastcgi_pass unix:/run/php-fpm/librenms.sock;
        fastcgi_split_path_info ^(.+\.php)(/.+)$;
        include fastcgi.conf;
    }
    location ~ /\.(?!well-known).* { deny all; }
}
```

```
sudo nginx -t
sudo systemctl enable nginx && sudo systemctl start nginx
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
```

4.8 Configure SNMP on Topton

Replace the default snmpd.conf with a minimal configuration:

```
sudo tee /etc/snmp/snmpd.conf << 'EOF'
agentAddress udp:161
rocommunity public 192.168.1.0/24
syslocation Homelab - Topton N150
syscontact SecAdmin
EOF
sudo systemctl enable snmpd && sudo systemctl start snmpd
sudo firewall-cmd --permanent --add-service=snmp
sudo firewall-cmd --reload
```

4.9 Set Up Cron Jobs and Scheduler

```
sudo cp /opt/librenms/dist/librenms.cron /etc/cron.d/librenms
sudo cp /opt/librenms/dist/librenms-scheduler.service /etc/systemd/system/
sudo cp /opt/librenms/dist/librenms-scheduler.timer /etc/systemd/system/
```

```
sudo systemctl enable librenms-scheduler.timer && sudo systemctl start librenms-scheduler.timer
sudo cp /opt/librenms/misc/librenms.logrotate /etc/logrotate.d/librenms
```

4.10 Web Installer

Open a browser and navigate to http://192.168.1.217 (replace with your LibreNMS host IP).

- Step 1 — Pre-install checks should all show green
- Step 2 — Configure Database: Host: localhost, User: librenms, Password: your password, Database: librenms
- Step 3 — Click Build Database and wait for completion
- Step 4 — Create Admin User with username, password, and email
- Step 5 — Select update channel (Daily) and theme, then click Finish Install

Important — Manual .env fix:

If the installer cannot write the .env file, edit it manually. Ensure DB_ lines are not commented out (no leading #) and that INSTALL=true is removed:

```
sudo nano /opt/librenms/.env
sudo chown librenms:librenms /opt/librenms/.env
sudo chmod 660 /opt/librenms/.env
sudo systemctl restart php-fpm nginx
```

5. Adding Devices

5.1 Topton (Linux, SNMP)

```
sudo -u librenms /opt/librenms/lnms device:add 192.168.1.217 --v2c --community public
```

5.2 Windows Server (SNMP Setup Required)

Run the following on the Windows Server in an Administrator PowerShell session:

```
Add-WindowsFeature SNMP-Service
Set-Service -Name SNMP -StartupType Automatic
Start-Service SNMP
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\SNMP\Parameters\ValidCommunities" -Name "public" -Value 4 -Type DWord
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\SNMP\Parameters\PermittedManagers" -Name "1" -Value "192.168.1.217"
Restart-Service SNMP
```

```
New-NetFirewallRule -DisplayName "SNMP" -Direction Inbound -Protocol UDP -
LocalPort 161 -Action Allow
New-NetFirewallRule -DisplayName "Allow ICMPv4" -Direction Inbound -Protocol
ICMPv4 -IcmpType 8 -Action Allow
```

Then add from Topton terminal:

```
sudo -u librenms /opt/librenms/lnms device:add 192.168.1.10 --v2c --community
public
```

5.3 Router (Ping Only)

Consumer routers typically do not support SNMP. Add with force flag to bypass checks:

```
sudo -u librenms /opt/librenms/lnms device:add 192.168.1.1 --force
```

6. Troubleshooting

6.1 Poller: No module named pymysql

```
sudo dnf install python3-PyMySQL -y
```

6.2 Poller: No module named dotenv

```
sudo dnf install python3-dotenv -y
```

6.3 Device Not Appearing on Map

Run discovery and polling manually. First check device IDs:

```
sudo mysql -e "SELECT device_id, hostname, ip, status FROM librenms.devices;"
sudo -u librenms /opt/librenms/lnms device:discover <device_id>
sudo -u librenms /opt/librenms/poller-wrapper.py 16
```

6.4 Windows SNMP Not Responding

Verify the PermittedManagers registry key includes the LibreNMS host IP and that the SNMP service is running. Test SNMP reachability from Topton:

```
snmpwalk -v2c -c public 192.168.1.10 sysDescr
```

Expected response includes Windows hardware and OS version string. If timeout occurs, recheck the PermittedManagers key and firewall rule.

6.5 .env File Write Failure During Install

The web installer may fail to write .env if PHP-FPM is not running as the librenms user, or if file permissions are incorrect. Manually edit the .env file using the content shown in the installer dialog. Ensure DB_ lines are uncommented and INSTALL=true is removed.

7. Results

After successful deployment, the LibreNMS availability map displays all monitored devices with live status indicators:

Device	IP	Status	SNMP	Details Available
Topton N150 (Rocky Linux)	192.168.1.217	Up (green)	Full v2c	CPU, RAM, traffic, interfaces, uptime
TAROX (Windows Server 2025)	192.168.1.10	Up (green)	Full v2c	OS info, uptime, availability, events
EE Smart Hub Router	192.168.1.1	Down (ping blocked)	None	Ping response graph only

The poller runs automatically every 5 minutes via cron, checking all devices and updating RRD graphs. Each device detail page provides OS identification, uptime tracking, overall traffic graphs, CPU and memory graphs (where SNMP MIBs are available), per-interface statistics, and a timestamped event log.

8. Skills Demonstrated

- Rocky Linux 9 server administration and package management
- Nginx web server configuration and PHP-FPM integration
- MariaDB database administration and schema deployment
- SNMP protocol configuration on Linux (net-snmp) and Windows Server
- SELinux context management for web application deployment
- Python dependency management with dnf on Rocky Linux
- Windows Server PowerShell administration and firewall configuration
- Network monitoring and NMS platform deployment
- Troubleshooting multi-component application stacks

9. Related Projects

- HomeLab Overview — github.com/arturskaufmanis/HomeLab-Overview
- Hardware Repair and Upgrade — arturskaufmanis.github.io/Hardware-repair-and-upgrade-project/
- Suricata IDS deployment on Topton N150 (transparent bridge)
- Splunk SIEM with Sysmon integration on Windows Server 2025
- BloodHound CE Active Directory analysis
- PingCastle AD security assessment